



Republic of Rwanda
Ministry of Education



BASIC REINFORCED CONCRETE DESIGN (RCD)

BDCBD501

Perform Basic Reinforced Concrete Design

Competence

RQF Level: 5

Learning Hours



Credits: 12

Sector: Construction and Building Services

Trade: Building Construction

Module Type: Specific

Curriculum: CBSBDC5001-TVET CERTIFICATE V IN BUILDING CONSTRUCTION

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Purpose statement	This specific module describes the skills, knowledge and attitude to perform reinforced concrete works on construction site. It is intended to be pursued by learner at TVET level V for TVET Certificate IV in Building Construction. At the end of this module, the learner will be able to: Interpret structural details, identify structural members’ connections, determine forces acting on structural members and design concrete structure elements.					
Learning assumed to be in place	N/A					
Delivery modality	Training delivery		100%	Assessment		Total 100%
	Theoretical content		30%	Formative assessment	30%	50%
	Practical work:		70%		70%	
	➤ Group project and presentation	20%				
	➤ Individual project /Work	50%				
			Summative Assessment		50%	

Elements of Competency and Performance Criteria

Elements of competency	Performance criteria
1. Interpret structural details	1.1. Mechanical properties of materials are correctly identified according to the project specifications.
	1.2. Structural members' details are correctly interpreted according to their types
	1.3. Structural members section is correctly interpreted based on the structural details.
	1.4. Bar bending schedules are correctly interpreted according to the type and form of structural member
2. Identify structural members connections	2.1. Reinforced concrete members are correctly positioned according to the construction structural drawings
	2.2. Reinforced concrete members interconnections are adequately identified according to the construction drawings
	2.3. Structure's supports are adequately identified based on the type of connections
	2.4. Support reactions are correctly identified based on the type of connections

3. Determine forces acting on structural members	3.1. Load is correctly identified according to the structure usage
	3.2. Load distribution is properly determined based on structure usage
	3.3. Equilibrium between external and internal forces is correctly determined according to the type of structural members service
4. Design concrete structure elements	4.1. Internal forces are correctly determined based on applied loads.
	4.2. Reinforced concrete design methods are properly used according to the structural member to be designed.
	4.3. Serviceability of reinforced concrete design is adequately determined according to the structural member service
	4.4. Reinforced concrete beam is correctly designed according to applied loads
	4.5. Reinforced concrete slab is correctly designed according to applied loads
	4.6. Reinforced concrete column is correctly designed according to applied loads
	4.7. Reinforced concrete footing is correctly designed according to applied loads

Intended Knowledge, Skills and Attitude

Knowledge	Skills	Attitude
✓ Describe the mechanical properties of materials ✓ Describe the reinforced concrete Structural members' ✓ Identify the type of connections of structural members ✓ Explain the type of member supports ✓ Explain the type of actions force ✓ Identify the units of measurement of different items ✓ Identify reporting tools	✓ Perform technical drawing interpretation ✓ Perform the Bar bending schedule Interpretation ✓ Perform the Bar bending structure notes Interpretation ✓ Apply measurement skills ✓ Apply load design skills ✓ Perform data collection, recording, and calculation of structural members ✓ Design the reinforced concrete slab ✓ Design the reinforced concrete beam	✓ Reasonable assumption. ✓ Use creativity and innovation throughout the design works ✓ Demonstrate punctuality during the implementation of the design project ✓ Demonstrate resourcefulness in the new design trends ✓ Be attentive to detail in the calculation of forces, design of the structural members, and assessment of project specifications ✓ Demonstrate effective communication skills to

	✓ Design the reinforced concrete column ✓ Design the reinforced concrete footing	- stakeholders. ✓ Have negotiation capability in costing ✓ Manage time effectively to meet project timelines. ✓ Having teamwork and collaboration spirit ✓ Practice compliance with safety and environmental regulations
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Course content

Learning outcomes	At the end of the module the learner will be able to: 1. Interpret structural details 2. Identify structural members connections 3. Determine forces acting on structural members 4. Design concrete structure elements
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Learning Outcome 1: Interpret Structural Details	Learning hours: 20 Hours
Indicative content	
• Determination of properties of reinforced concrete ✓ Identification of mechanical properties • Strain • Stress • Stress-strain curve ✓ Identification of physical properties • Unit weight • Density ✓ Identification of geometric properties • Centre of gravity • First moment of area • Section modulus	

- Moment of inertia
 - Radius of gyration
- ✓ Calculation of geometrical and mechanical properties of reinforced concrete member
- **Interpretation of structural member sections**
 - ✓ Slab sectioning
 - Transversal section
 - Longitudinal section
 - Slab section labelling
 - Slab section drawing details
 - ✓ Beam sectioning
 - Transversal section
 - Longitudinal section
 - Beam section labelling
 - Beam section drawing details
 - ✓ Column sectioning
 - Transversal section
 - Longitudinal section
 - Column section labelling
 - Column section details
 - ✓ Foundation sectioning
 - Transversal section
 - Longitudinal section
 - Foundation section labelling
 - Foundation section details
 - ✓ A typical reinforced concrete building sectioning
 - Transversal section
 - Longitudinal section
 - A typical reinforced concrete building labelling
 - A typical reinforced concrete building section details

- **Preparation of bar bending schedule**

- ✓ Plan for bending schedule works
 - Analysing the structural design reinforcement details
 - Determining the steel quantity take off
 - Arranging the steel bars
 - Assigning bar marks
 - Creating the bar bending schedule
- ✓ Prepare bar bending schedule for slabs
 - Steel bar for one way slab
 - Steel bar for two way slab
- ✓ Prepare bar bending schedule for beams
 - Steel bar for simple beams
 - Steel bar for continuous beams
- ✓ Prepare bar bending schedule for columns
 - Calculation of the length of vertical bar
 - Calculation of lapping length
 - Cutting length of stirrups
 - Number of stirrups
- ✓ Prepare bar bending schedule for footings
 - Steel bar for plain footings
 - Steel bar for slopped footings
 - Steel bar for combined footings

Resources required for the indicative content

Equipment	• Computer, printer, projector
Materials	• Papers, pencil, pens, marker
Tools	• Calculator, mathematical set
Facilitation techniques	Group discussion, scenario, simulation, individual activity • Trainer asks the students in manageable groups to discuss on the mechanical properties

	of reinforced concrete, physical properties and geometrical properties. • Trainer prepares slab structural sections in manageable groups. He/She manage a group discussion on interpretation of structural member sections and he/she asks the students to discuss on preparation of bar bending schedule. • Trainer prepares a bar bending schedule and He/ She asks student to interpret individually a bar bending schedule.
Formative assessment methods	• Written assessment • Oral assessment • Performance assessment

Learning outcome 2: Analyze Structural Members Connections		Learning hours: 20 Hours
Indicative content		
• Identification of reinforced concrete members interconnections ✓ Slab-beam connection ✓ Beam-column ✓ Column-footing ✓ Elaboration of structural connections • Identification of reinforced concrete supports ✓ Roller Supports ✓ Hinged/Pinned Supports ✓ Fixed Supports ✓ Simply Supports (A Roller support at one end of the member (beam) and a Hinged/Pinned support at the other end of the member (beam)) ✓ Elaborate free body diagram of structural system • Calculation of reinforced concrete support reactions ✓ Vertical reaction or Horizontal reaction for Roller supports ✓ Vertical reaction and Horizontal reaction for Hinged/Pinned supports ✓ Vertical reaction, horizontal reaction and moment/Rotation for Fixed supports		
Resources required for the indicative content		
Equipment	• Printer, computer, projector	

Materials	Papers, pencil, pens, marker
Tools	Calculator, mathematical set
Facilitation techniques	Group discussion, scenario, simulation, individual activity - Trainer discuss on the reinforced concrete members interconnections and He/ ask student to summarize the main structural connections. - Trainer discuss on the reinforced concrete supports and He/ ask student to summarize the main reinforced concrete supports. - Trainer prepares a summary on reinforced concrete supports and He/She asks the students to sketch free body diagram of structural system individually. - Trainer prepares exercises on supports reactions. He/ She asks student to calculate individually support reactions. - Trainer demonstrate to the students by calculating reinforced concrete support reactions and asks the students to follow the procedures.
Formative assessment methods	- Written assessment - Oral assessment - Performance assessment

Learning Outcome 3: Determine Forces Acting on Structural Members	Learning hours: 30 Hours
Indicative content	
Determination of loads in reinforced concrete members ✓ Identification of loads based on their nature and sources - Dead loads - Live loads (Imposed loads) - Environmental loads ✓ Identification of loads based on structural idealisation - Point load - Uniformly distributed load - Uniformly varying load - Load combinations ✓ Calculation of loads in reinforced concrete members - Dead loads - Live loads (Imposed loads)	

- **Applying the calculated load on structural members**
 - ✓ Slabs
 - One way slab
 - Two way slab
 - ✓ Beams
 - Simple beams
 - Continuous beams
 - ✓ Columns
 - ✓ Footings
 - ✓ Typical load distribution on the entire building
- **Determining the equilibrium between external and internal forces**
 - ✓ Newton's third law
 - ✓ Analysing the external forces (applied loads)
 - ✓ Computation of support reactions
 - ✓ Calculation of internal forces

Resources required for the indicative content

Equipment	• Printer, computer, projector
Materials	• Papers, pencil, pens, marker
Tools	• Calculator, mathematical set
Facilitation techniques	Group discussion, scenario, simulation, individual activity • Trainer asks the students to discuss on reinforced concrete members interconnections and He/ asks student to discuss the main structural connections. • Trainer prepares a summary on reinforced concrete supports and He/She asks the students to sketch free body diagram of structural system individually. • Trainer prepares exercises on supports reactions. He/ She asks student to calculate individually support reactions. • Trainer demonstrate to the students how to determine the equilibrium between external and internal forces and asks the students to follow the procedures.
Formative assessment methods	• Written assessment • Oral assessment • Performance assessment

Learning outcome 4: Design reinforced concrete structural elements

Learning hours: 50 Hours

Indicative content

- **Determination of internal forces**
 - ✓ Identification of Types of internal forces
 - Axial forces
 - Shear forces
 - Bending moments
 - ✓ Calculation of internal forces
- **Analysing the methods of design in reinforced concrete structures**
 - ✓ Permissible stress method
 - ✓ Load factor method
 - ✓ Limit state method
 - Ultimate limit state
 - Serviceability limit state
- **Determination of reinforced concrete design for serviceability**
 - ✓ Cover to concrete
 - Nominal cover
 - Conditions of exposure
 - ✓ Deflection control
 - ✓ Cracking control
 - ✓ Maximum clear spacing of reinforcement
 - Beams
 - Slabs
 - ✓ Minimum spacing of reinforcement

- ✓ Minimum area of reinforcement
- ✓ Maximum area of reinforcement
- ✓ Reinforcement detailing
- **Design reinforced concrete beam**
 - ✓ Analysis of a section
 - Failure modes
 - Analyse dimensions of the section (Neutral axis, width, effective depth, lever arm)
 - Location of resultant forces in compression and tension zones
 - ✓ Design approach
 - Estimate the cross section dimensions of the beam
 - Determine the design load (Ultimate load)
 - Determine the bending moment
 - Determine the ratio of the depth of the elastic neutral axis to the overall depth beam (k)
 - Check the value of (k) against the limiting value (k')
 - Determine the lever arm ratio (lever arm (z)/effective depth (d))
 - Check the lever arm ratio (z/d) against the minimum ($z=0.775d$) and the maximum value ($z=0.95d$)
 - Determine of lever arm (z)
 - Calculate the area of longitudinal reinforcement (A_s)
 - Select the reinforcement from the Table
 - Provide the reinforcement area slightly greater than the calculated steel area
 - ✓ Prepare reinforcement detailing
 - Draw the cross section area the beam with all dimensions
 - Provide the size of the longitudinal reinforcement
- **Design reinforced concrete slab**
 - ✓ Identification load types and magnitudes for design
 - Dead loads (Self-weight of the slab and permanent fixtures)
 - Live loads (Load due to the use of the building + temporary fixtures)
 - Slab type (One way and two way)

- ✓ Identification of slab panel types and associated support conditions
 - Interior panels (Supported on four edges)
 - One short edge discontinuous
 - Two adjacent edges discontinuous
 - One long edge discontinuous
- ✓ Calculate the ratio between the slab dimensions
- ✓ Determine the span between supports and overall dimensions of the slab.
 - Check the ratio between the longer side and the shorter side
 - Slab preliminary thickness and depth
 - Estimate the slab thickness
 - Assume the initial diameter of steel reinforcement
 - Determine the effective depth of the slab
- ✓ Design approach
 - Estimate the cross section dimensions of the slab
 - Determine the design load (Ultimate load)
 - Determine the bending moment
 - Determine the ratio of the depth of the elastic neutral axis to the overall depth beam (k)
 - Check the value of (k) against the limiting value ($k'=0.156$)
 - Determine the lever arm ratio (lever arm (z)/effective depth (d))
 - Check the lever arm ratio (z/d) against the minimum ($z=0.775d$) and the maximum value ($z=0.95d$)
 - Determine lever arm (z)
 - Calculate the area of longitudinal reinforcement (A_s)
 - Select the reinforcement from the Table
 - Provide the reinforcement area slightly greater than the calculated steel area
 - Determine the spacing between reinforcement
 - Check deflection (Short span/Effective depth)
 - The deflection should not exceed the provided value
- ✓ **Prepare reinforcement detailing**

- Draw the top view of the slab with all dimensions
- Provide the size of the longitudinal reinforcement and the spacing
- **Designing a reinforced concrete column**
 - ✓ Identification of column
 - Column sections
 - Failure modes of columns
 - Slenderness ratio
 - Short and slender columns
 - Braced and unbraced columns
 - Internal column
 - External column (Edge columns, Corner columns)
 - ✓ Types of short columns based on load distribution
 - Columns resisting axial load only
 - Columns supporting an approximately symmetrical arrangement of beams
 - ✓ Design approach for short columns
 - Estimate the cross section dimensions of the beam based on the span and the applied load
 - Determine the design load (Ultimate load)
 - Calculate the area of longitudinal reinforcement (A_{sc})
 - Select the reinforcement from the Table
 - Provide the reinforcement area slightly greater than the calculated steel area
 - ✓ Link/stirrups design
 - Critical zone
 - Non-critical zone
 - Calculate the transverse reinforcement (A_{scv})
 - Select the reinforcement from the Table
 - Provided reinforcement area slightly greater than the calculated steel area
 - Spacing estimation (In both critical and non-critical zones)
 - ✓ Reinforcement detailing
 - Draw the cross section of the column with all dimensions

- Provide the size of the longitudinal reinforcement and the transversal reinforcement with the cross section
- Draw the longitudinal section of the column with all dimensions
- Provide the size of longitudinal reinforcement and the transversal reinforcement (In both critical and non-critical zones) with the calculated spacings
- **Designing a reinforced concrete isolated footing**
 - ✓ Importance of foundation
 - ✓ Identification of Types of footings
 - Plain/Single footing
 - Slopped footing
 - Stepped footing
 - Shoe/Eccentric footing
 - Combined footing
 - ✓ Identification of Bearing capacity
 - Ultimate bearing capacity
 - Safe bearing capacity
 - Permissible bearing pressure
 - Calculate ultimate bearing capacity
 - ✓ Soil pressure distribution under footing
 - ✓ Blinding concrete
 - ✓ Calculation of bearing pressure
 - Pressure when the bending moment is zero ($M = 0$)
 - Pressure when the eccentricity (e) is less or equal to effective depth divided by 6 ($e \leq D/6$)
 - Pressure when the eccentricity (e) is greater than the effective depth divided by 6 ($e > D/6$)
 - ✓ Design approach – Bending moment
 - Soil bearing pressure is determined from soil tests
 - Determine the characteristic load
 - Estimate the square footing area
 - Calculate the required minimum size of a square footing

- Provide the size of a square footing slightly greater than the calculated square footing area
- Calculate the real pressure of soil
- Check the real pressure of soil against the given soil bearing pressure
- Determine the ultimate bearing pressure
- Determine the maximum bending moment
- Assume the overall depth (h) of the footing
- Determine the effective depth of the footing
- Determine the ratio of the depth of the elastic neutral axis to the overall depth beam (k)
- Check the value of (k) against the limiting value ($k' = 0.156$)
- Determine the lever arm ratio (lever arm (z)/effective depth (d))
- Check the lever arm ratio (z/d) against the minimum ($z = 0.775d$) and the maximum value ($z = 0.95d$)
- Determine of lever arm (z)
- Calculate the area of longitudinal reinforcement (A_s)
- Select the reinforcement from the Table
- Provide the reinforcement area slightly greater than the calculated steel area
- Determine the reinforcement distribution
- Determine the spacing between reinforcement
- ✓ Reinforcement detailing
 - Draw the top view of the footing with all dimensions
 - Provide the size of the longitudinal reinforcement (in both size) and the spacing
- ✓ Design approach – Punching shear
 - Calculate the perimeter around the column
 - Calculate the design shear stress (ν) around the perimeter of the column
 - Check if the value of shear stress (ν) is less than **$0.8 \sqrt{f_{cu}}$ and/or $5N/mm^2$**
 - The overall depth (h) of the footing is adequate if stress (ν) is less than **$0.8 \sqrt{f_{cu}}$ and/or $5N/mm^2$**
 - Determine the concrete shear capacity (ν_c)
 - Calculate the perimeter at 1.5d distance from the face of the column

- Calculate the design shear stress (v) at 1.5d distance from the face of the column
- If $v \leq v_c$, the design is satisfactory (free from punching shear)

Resources required for the indicative content

Equipment	• Computer, printer, projector
Materials	• Paper, pencil, pens
Tools	• Calculator, mathematical set
Facilitation techniques	Group discussion, scenario, simulation, individual activity • Trainer asks the students to discuss on different types of internal forces. He /She asks students on He/ she asks student to presents their findings. • Trainer prepares an exercise on how internal forces are spread throughout the concrete member and He/she asks student to calculate internal reactions individually. • Trainer presents different design methods of analyzing reinforced concrete structures. In manageable groups, He/she asks the students to discuss on each method of analysis. • Trainer prepares a discussion on how to determine reinforced concrete design for limit of serviceability. In manageable groups He/ She asks student to discuss about reinforced concrete design for limit of serviceability. • Trainer prepares a sketch of beam and presents design approach. He /She asks students on He/ she asks student to discuss on beam design approach. • Trainer prepares a sketch of column and presents design approach. He /She asks students on He/ she asks student to discuss on column design approach. • The trainer sketches a footing and introduces their design approach. They then encourage the students to discuss the footing design approach.

Integrated/Summative assessment (For specific module)

Integrated situation

HID Ltd., a Kigali-based consulting firm, has been contracted to review the structural design documents prepared by another company for a new building project. Our role is to ensure the stability and structural integrity of the building based on our engineering expertise.

The review will focus specifically on the one-way reinforced concrete slab and footing designs. We will provide a detailed report outlining our assessment and any necessary recommendations.

To facilitate this review, please provide the following specifications and data:

List of Specifications and Data Here:

- Slab type: Simply supported one-way slab.
- Span (L): 600 mm.
- Cover :25 mm
- Width (b): 300 mm
- Live load: 2.4 kN/m²
- Slab thickness (h) = 150mm
- Concrete strength (fc'): 25 MPa.
- Steel yield strength (fy): 460 MPa.
- Design steel tensile strength (Rs): 40N/m2
- My+ Positive bending moment on slab: 13.05 KNm
- My- Negative bending moment on slab: 31.23 KNm
- Take : $Z/d = 0.95$
- The deflection should not exceed: 32.3
- A column 220 mm x 220 mm carries a dead load of 590697 N and an imposed load of 225000 N. The safe bearing pressure is 200 kN/m². Design a square base to resist the loads. The concrete grade is 37 and the reinforcement grade 460. The condition of exposure is moderate and the nominal cover is 50 mm for concrete cast against blinding.

For Slab you are requested to:

1. calculate dead Load
2. calculate total Load
3. calculate bending moment (M)
4. Check for deflection
5. Determine area of reinforcement
6. Provide the size of the longitudinal reinforcement and the spacing
7. Provide reinforcement detailing

Table 31. Reinforcement areas (mm²)

No	Bar diameter (mm)							
	8	10	12	16	20	25	32	40
1	50.3	78.5	113.1	201.1	314.2	490.9	804.2	1256.6
2	101	157	226	402	628	982	1608	2513
3	151	236	339	603	942	1473	2413	3770
4	201	314	452	804	1257	1963	3217	5027
5	251	393	565	1005	1571	2454	4021	6283
6	302	471	679	1206	1885	2945	4825	7540
7	352	550	792	1407	2199	3436	5630	8796
8	402	628	905	1608	2513	3927	6434	10053
9	452	707	1018	1810	2827	4418	7238	11310
10	503	785	1131	2011	3142	4909	8042	12566
11	553	864	1244	2212	3456	5400	8847	13823
12	603	942	1357	2413	3770	5890	9651	15080
13	653	1021	1470	2614	4084	6381	10455	16336
14	704	1100	1583	2815	4398	6872	11259	17593
15	754	1178	1696	3016	4712	7363	12064	18850
16	804	1257	1810	3217	5027	7854	12868	20106
17	855	1335	1923	3418	5341	8345	13672	21363
18	905	1414	2036	3619	5655	8836	14476	22619
19	955	1492	2149	3820	5969	9327	15281	23876
20	1005	1571	2262	4021	6283	9817	16085	23876
21	1056	1649	2375	4222	6597	10308	16889	26389
22	1106	1728	2488	4423	6912	10799	17693	27646
23	1156	1806	2601	4624	7226	11290	18498	28903
24	1206	1885	2714	4825	7540	11781	19302	30159
25	1257	1963	2827	5027	7854	12272	20106	31416
26	1307	2042	2941	5228	8168	12763	20910	32673
27	1357	2121	3054	5429	8482	13254	21715	33929
28	1407	2199	3167	5630	8796	13744	22519	35186
29	1458	2278	3280	5831	9111	14235	23323	36442
30	1508	2356	3393	6032	9425	14726	24127	37699
31	1558	2435	3506	6233	9739	15217	24932	38956
32	1608	2513	3619	6434	10053	15708	25736	40212
33	1659	2592	3732	6635	10367	16199	26540	41469
34	1709	2670	3845	6836	10681	16690	27344	42726
35	1759	2749	3958	7037	10996	17181	28149	43982
36	1810	2827	4072	7238	11310	17671	28953	45239
37	1860	2906	4185	7439	11624	18162	29757	46496
38	1910	2985	4298	7640	11938	18653	30561	47752
39	1960	3063	4411	7841	12252	19144	31366	49009
40	2011	3142	4524	8042	12566	19635	32170	50265

For footing you are requested to:

1. Calculate design load:
2. Calculate final dead load
Assume the self-weight of the footing and the soil above to be **10%** of the total characteristic load,
3. Calculate the final characteristic load N_{Ed}
4. Calculate the area of the footing
5. Determine the Required minimum size of the footing =
6. Calculate the real pressure
7. Determine the ultimate design load
8. Calculate the ultimate bearing pressure

9. Calculation the maximum bending moment
10. Determine the effective depth
Assume main reinforcement as 16 mm diameter bars.
11. Calculate the area of reinforcement required
12. Determine the distribution of reinforcement
13. Calculate the ultimate shear force
14. Design shear stress
15. Prepare reinforcement detailing

Resources

Tools	Calculator, mathematical set
Equipment	Computer, printer, projector
Materials/ Consumables	Pens, papers, pencil

Assessable outcomes	Assessment criteria (Based on performance criteria)	Indicator	Observation		Marks allocation
			Yes	No	
Learning outcome 1: Interpret structural details	1.1. Mechanical properties of materials are correctly identified according to the project specifications.	Mechanical properties are identified.			2
		Physical properties are identified			1
		Geometric properties are identified.			2
	1.2. Structural members' details are correctly interpreted according to their types	Structural members' details are interpreted			2
	1.3. Structural members section is correctly interpreted based on the structural details.	Slab section is interpreted			1
		Beam section is interpreted			2
		Column section is interpreted			2
		Footing section is interpreted			2
	1.4 Bar bending schedules are correctly interpreted according to the type and form of structural member	BBS is interpreted			2
Learning	2.1 Structure supports are	Concrete members			2

outcome 2: Identify structural members connections	adequately identified based on the type of connections	interconnections are identified			
		Reinforced concrete supports are identified			2
		Support reactions are calculated			2
	2.2 Reinforced concrete members are correctly positioned according to the construction structural drawings	Reinforced concrete members positions are shown			2
	2.3. Reinforced concrete members interconnections are adequately identified according to the construction drawings	Concrete members interconnections are identified			2
Learning outcome 3: Determine forces acting on structural members	3.1. Stability of the structure is adequately identified according to the mode of loading and bearing capacity	Load nature is identified			2
		Structural load idealization is identified			2
	3.2. Load is correctly identified according to the structure usage	Dead load are calculated			1
		Live load are applied			1
	3.3. Load distribution is properly determined based on structure usage	Load are applied			2
	3.4. Actions forces are correctly identified according to the type of structural members service	Action forces are identified			2
Learning outcome 4: Design concrete structure elements	4.1 Internal forces are correctly determined based on applied loads.	Axial forces are calculated			2
		Shear forces are calculated			2
		Bending moment is determined			2
		Analyzing the methods is identified			2
		Design for serviceability is determined			2
	4.2. Reinforced concrete slab is correctly designed according to applied loads	Load types and magnitudes are identified			2
		Slab panel types and associated support conditions are identified			2

		Ratio between the slab dimensions is determined			2
		Design approach is chosen			2
		Concrete cover is determined			2
		Deflection is controlled			2
		Maximum clear spacing of reinforcement is determined			2
		Minimum spacing of reinforcement is determined			2
		BottomOutcome area of reinforcement is calculated			2
		Top area of reinforcement is calculated			2
		Reinforcement detailing is prepared			2
	4.3. Reinforced concrete beam is correctly designed according to applied loads	Section is analyzed			2
Design approach is determined				2	
Reinforcement detailing is prepared				2	
	4.4. Reinforced concrete column is correctly designed according to applied loads	Column is identified			2
		Column load is identified			2
		Column design approach is selected			2

		Link/stirrups are designed			2
		Reinforcement detailing is prepared			2
	4.5 Reinforced concrete footing is correctly designed according to applied loads	Importance of foundation is explained			2
		Types of footings are identified			2
		Bearing capacity is provided			2
		Soil pressure distribution under footing is calculated			2
		Bearing pressure is calculated			2
		Design approach is selected			2
	Reinforcement detailing is prepared			2	
	Punching shear is calculated			2	
Total marks		100			
Percentage Weightage		100%			
Minimum Passing line % (Aggregate): 70%					

References

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